

# SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 17:45 UTC

## SEC P vs NP — notebook summary

Generated 2026-05-18 17:45 UTC

- Cycles recorded: **787**
- Time span: 596.1h (~1.32 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp\_notebook.jsonl

### Verdict distribution

| Verdict      | Count |
|--------------|-------|
| INCONCLUSIVE | 747   |
| BARRIER_HIT  | 20    |
| FALSIFIED    | 16    |
| SUPPORTED    | 4     |

### Mathematical fields explored (field\_A)

| Field                                                                             | Cycles |
|-----------------------------------------------------------------------------------|--------|
| Real Algebraic Geometry                                                           | 40     |
| Schur-Weyl Duality                                                                | 28     |
| Free Probability Theory                                                           | 19     |
| Representation Theory of Symmetric Groups                                         | 18     |
| Matroid Theory                                                                    | 16     |
| Free Probability                                                                  | 14     |
| Additive Combinatorics                                                            | 13     |
| Noncommutative $L^p$ Geometry                                                     | 10     |
| Random Matrix Theory                                                              | 8      |
| Polymatroid Theory                                                                | 8      |
| Algebraic Geometry                                                                | 7      |
| Spectral Graph Theory                                                             | 7      |
| Finite Geometry                                                                   | 6      |
| Plethysm Theory                                                                   | 6      |
| Persistent Homology                                                               | 5      |
| Noncommutative Harmonic Analysis                                                  | 5      |
| Ergodic Circuit Framework (communication complexity via dynamical systems)        | 4      |
| {“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”} | 4      |
| Fourier Analysis on Boolean Functions                                             | 4      |

| Field                                                                                              | Cycles |
|----------------------------------------------------------------------------------------------------|--------|
| Algebraic Geometry over Finite Fields                                                              | 4      |
| Fourier Analysis of Boolean Functions                                                              | 4      |
| Algebraic Geometry of Secant Varieties                                                             | 4      |
| FOURIER_ANALYSIS                                                                                   | 4      |
| Noncommutative Geometry                                                                            | 4      |
| Fourier analysis of boolean functions                                                              | 3      |
| Algebraic Topology                                                                                 | 3      |
| matroid theory                                                                                     | 3      |
| Coarse Geometric Lifting (CGL) — Geometric Complexity Theory                                       | 3      |
| {“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”} | 3      |
| Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)                                              | 3      |

### Last 15 cycles (chronological)

| Time                 | Verdict      | Title                                                                            |
|----------------------|--------------|----------------------------------------------------------------------------------|
| 2026-05-18 08:28 UTC | INCONCLUSIVE | Halász $L^2$ Spectrum Discrepancy Lower-Bounds                                   |
| 2026-05-18 08:56 UTC | INCONCLUSIVE | Sign-Matrix Rigidity Mostar Index of Gate-Adjacency Graph Bounds                 |
| 2026-05-18 09:23 UTC | INCONCLUSIVE | $ACC^0[m]$ Size for MOD Patience-Sort LIS of XOR-Lifted Row Permutations         |
| 2026-05-18 09:54 UTC | INCONCLUSIVE | Bounds $CC^D$ Effective Resistance of Charged Vertex Pairs Bounds                |
| 2026-05-18 10:26 UTC | INCONCLUSIVE | Tseitin DPLL $p=3$ Path-Family Modulus Lower-Bounds Tseitin                      |
| 2026-05-18 11:02 UTC | INCONCLUSIVE | Tree-Resolution Viennot Heap Radius of Clause-Conflict Graph Bounds              |
| 2026-05-18 11:29 UTC | INCONCLUSIVE | Frege Depth Lee-Yang Zero Cluster Angle of Cut Polynomial Bounds                 |
| 2026-05-18 11:55 UTC | INCONCLUSIVE | Max-Cut SoS- Additive Energy of Out-Degree Sequence Bounds $ACC^0[2]$            |
| 2026-05-18 12:32 UTC | INCONCLUSIVE | MOD <sub>3</sub> Size SBM Detectability Gap of Literal Conflict Graph Bounds     |
| 2026-05-18 13:00 UTC | INCONCLUSIVE | DPLL Time Fiedler Participation Entropy Lower-Bounds Tseitin                     |
| 2026-05-18 13:47 UTC | INCONCLUSIVE | Tree-Resolutio Hypercontractive Term-Pair Contraction Defect Bounds Monotone CLI |

| Time                 | Verdict      | Title                                                                                      |
|----------------------|--------------|--------------------------------------------------------------------------------------------|
| 2026-05-18 16:26 UTC | INCONCLUSIVE | Lovász-Theta of Term-Conflict Graph Bounds Monotone k-CLIQUE DNF                           |
| 2026-05-18 17:00 UTC | INCONCLUSIVE | Sidon B <sub>2</sub> -Witness Failure of ACC <sup>0</sup> [2] for Sipser via MOD-Gate Outp |
| 2026-05-18 17:28 UTC | INCONCLUSIVE | RSK Shape Concentration Separates Permanent from Padded Determina                          |
| 2026-05-18 17:45 UTC | INCONCLUSIVE | Tusnady 2-Box Discrepancy of Clause-Polarity Cloud Bounds DPLL Si                          |

## How to read the reports

- reports/supported\_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified\_findings.md — FALSIFIED with counterexamples.
- reports/daily\_YYYY-MM-DD.md — chronological log by day.
- pvsnp\_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

## PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```